

Alexis Bollack

ROBOTICS ENGINEERING STUDENT · EPFL

Paris, France and Lausanne, Switzerland

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Summary

Robotics Master's student at EPFL (double degree with Ecole Centrale de Nantes) specializing in the intersection of Reinforcement Learning and autonomous systems. Hands-on experience in ROS2, robot control, and RL agents training for complex locomotion and manipulation using Isaac Lab and MuJoCo Warp. Strong foundation in dynamics, control systems, and Deep Learning. Seeking a **6-month industry internship** in robotics starting in September 2026.

Experiences

Mathis Group

Geneva, Switzerland

SEMESTER PROJECT

Feb. 2026 – June. 2026

Research laboratory of EPFL specialising in computational neuroscience and machine learning.

- Trained RL agent to play table tennis using complex musculoskeletal models within the MyoSuite and MuJoCo frameworks
- Implemented and built upon research paper methodologies to achieve SOTA performance.
- Transition from the MuJoCo implementation to MuJoCo Warp, enabling parallel simulation on GPU using the mjlabs framework.

EPFL IA Team, Paws project

Lausanne, Switzerland

RL ENGINEER

Feb. 2026 – June. 2026

- Developed quadruped locomotion with reinforcement learning using Isaac Lab.
- Worked on sim2sim and sim2real gap to implement the policy on the unitree go2 quadruped robot.

Institute of Intelligent Systems and Robotics (ISIR)

Paris, France

RESEARCH INTERN – PAID INTERNSHIP

Apr. 2025 – Aug. 2025

Research laboratory of Sorbonne University / CNRS specialising in robot learning and human-robot interaction.

- Developed IGM/DGM geometric models for a parallel robot and validated them against physical measurements.
- Designed a GUI for trajectory planning and execution.
- Implemented ROS2 low-level motor control over CAN bus.
- Integrated the robot with a motion capture system for closed-loop position feedback.

Omni

Paris, France

INTERN – PAID INTERNSHIP

Jun. 2024 – Jul. 2024

Start-up dedicated to the mobility of disabled people using electric devices.

- Preparation, repair and reconditioning of electric scooters.
- Created educational and technical videos for customers and technicians.

Robotics and AI Projects

2026 **Hackathon**, RoboHack2026 (EPFL): Program a robotic guide dog with GPS-based global navigation and agentic human interaction, code name Northern-Star

Lausanne,
Switzerland

2024 **Group Leader**, Naval Group, Aquabot challenge
Led a team of 5 in the Aquabot 2024 autonomous boat challenge.
Programmed ROS2 nodes for navigating an autonomous boat in swell and wind-driven environments.

Nantes, France

2024 **Hackathon**, Hugging Face LeRobot, reinforcement learning on a low-cost robotic arm

Toulouse, France

2024 **Competition**, Sumo Bot, built a differential-wheeled autonomous robot to navigate an arena and engage opponents

Nantes, France

2024 **Certificate**, Coursera Machine Learning Specialisation: Machine learning and deep learning

Online

2023 **Engineering Project**, Bicycle anti-lock braking system
Conducted a feasibility study of an ABS system applied to bicycles; built a working prototype using an Arduino and 3D-printed parts
Collaborated with a senior e-VTT engineer from Décathlon to validate design choices.

Paris, France

Education

EPFL | École Polytechnique Fédérale de Lausanne

Switzerland

MSC IN ROBOTICS (DOUBLE DEGREE)

2025 – present

- Specializing in the intersection of Reinforcement Learning and autonomous systems.
- Focusing on Robot Dynamics, Control Systems, and Deep Learning.
- Relevant courses: Legged robots; Reinforcement learning; Machine learning I; Model predictive control; Advanced control systems; Applied data analysis; Deep learning.

École Centrale de Nantes

France

MSC IN ENGINEERING

2023 – 2025

- Engineering Student at a highly selective French *Grande École* conferring a degree equivalent to a Master of Science.
- Specialization in Robotics, Control Systems, and Software Engineering.

Lycée Chaptal

Paris, France

CLASSES PRÉPARATOIRES AUX GRANDES ÉCOLES (MPSI/MP)

2021 – 2023

- Two-year intensive undergraduate program in Mathematics and Physics preparing for competitive national exams.

Lycée Carnot

Paris, France

BACCALAURÉAT (MATHS-PHYSICS) WITH HONOURS

June 2021

- Equivalent to High School Diploma.

Technical Skills

Simulation & Learning	MuJoCo, Isaac Lab, RSL-RL
Programming	C++, Python, MATLAB, Git
Robotics	ROS2
CAD	Fusion 360, Solidworks

Languages

French	Native speaker
English	Fluent (TOEIC 960/990, TOEFL 100/120)
Spanish	Intermediate

Sports

Volleyball	Lausanne Volleyball Club, M2 team
Half Marathon	Running
Table Tennis	French championship (-13 yrs)